ANDROID PROGRAMMING PRACTICAL 11

Module 2 · SQLite with ListView or RecyclerView

Aim

Integrate SQLite data with ListView or RecyclerView.

Project name: SQLiteListDemo

Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

Procedure

1. Create a SQLiteOpenHelper class with a table and methods to insert and query records.

2. Add a ListView or RecyclerView to the activity layout and provide a form to add records.

3. Query the database into a list of model objects or strings.

4. Create an ArrayAdapter for ListView or a RecyclerView.Adapter for RecyclerView.

5. Refresh the adapter after insert, update, or delete operations.

6. Handle empty database results with a useful empty-state message.

7. Run the app, add several records, restart the app, and confirm persisted records are displayed.

Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

Expected result

SQLite records appear in the selected list component and persist after restarting the app.

Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

Conclusion

The stated Android concept was implemented and verified through the steps and tests above.